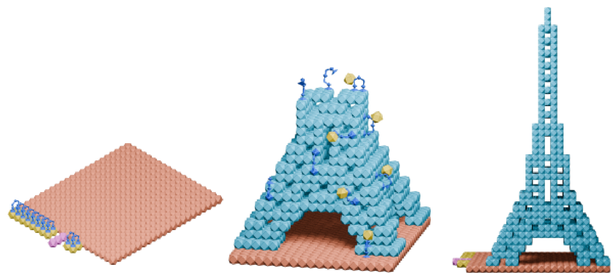
Conflict-Free Distributed Pathfinding for Collective Construction Using a Roadmap

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Introduction

- NASA's Automated Reconfigurable Mission Adaptive Digital Assembly Systems (ARMADAS) project aims to autonomously assemble structures from building blocks, using robots that traverse the structure [1].
- **Current limitation:** Centrally planning robot paths across a growing structure creates a computational bottleneck that limits scalability [2].
- **Research question:** Can agents plan paths such that their trajectories are collision-free with fully distributed computation and no negotiation, prioritization, or task-swapping?



An example structure the ARMADAS robotic system can build [1, 2].

Problem statement

- **Objective 1:** distributed multi-agent pathfinding for collective construction with conflict-free trajectories, without negotiation, prioritization, or task-swapping.
- **Objective 2:** multi-agent navigation where agents make progress towards their goals in congested environments.

Domain model

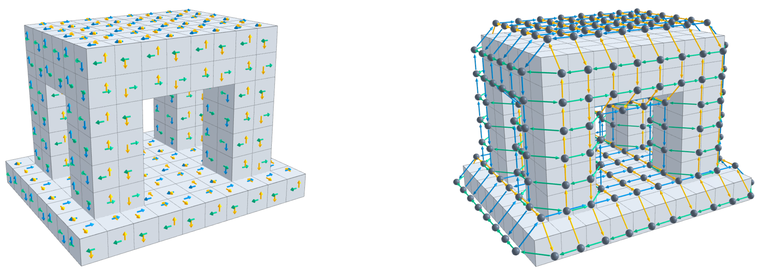
Robots build structures from modular blocks that snap together, retrieve a block from the depot, and navigate to its delivery location across the structure. For now, the structure remains fixed; deliveries do not modify it.

Robot model

- Occupy one empty grid cell face-adjacent to an exposed block.
- Slide across edge-adjacent exposed faces, including around corners [3].
- Never move into an occupied cell.

Conflict-Free Directed Roadmap

Arrows drawn on a structure's surface indicate where robots can move. Each arrow's time phase (color) indicates when that edge is traversable.



Left: a polycube with its Conflict-Free Directed Roadmap. Right: the resulting state-transition graph.

Solves Objective 1: distributed pathfinding with collision-free trajectories.

Guarantees

- **Conflict-free.** Edge conflicts are impossible because the graph is oriented; vertex conflicts are impossible because no two incoming arrows are active at once.
- **Strongly connected.** A path for a robot exists between any two cells in the graph.
- **Deterministic and stable.** Arrows depend only on a face's coordinates and orientation, so the roadmap over unchanged regions never shifts as the structure grows or shrinks.

Does not guarantee that robots will reach their destination when there is traffic.

Policies for navigating a congested roadmap

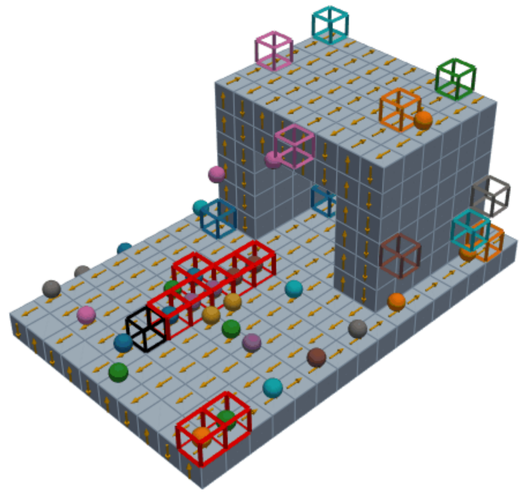
Robot policies address Objective 2. Robots plan paths through the roadmap with A*. When the next cell in a robot's path is occupied, a **policy** decides what to do:

- **Wait:** hold position until the cell is vacant and the arrow into it is active.
- **Re-plan:** add +1 penalty to the blocked cell and re-plan; wipe penalties after moving.
- **Remember:** *re-plan*, but penalties persist and decay over time.
- **Congestion:** *remember*, but penalize all occupied, adjacent cells.

Experimental setup

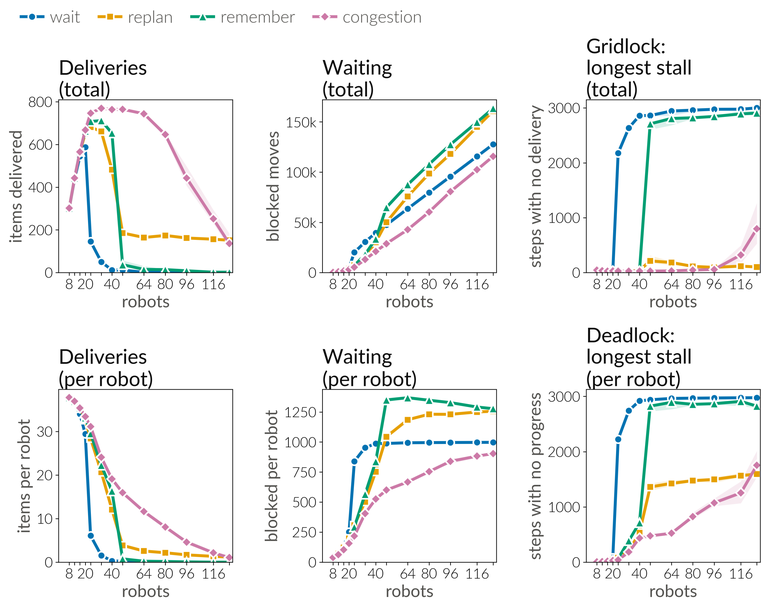
Multi-Agent Pickup and Delivery on a static structure with 412 reachable cells. Unlike collective construction, the structure is not modified; this is an informative preliminary step for validating a static, conflict-free roadmap before a dynamic one.

- Robots without an item go to the depot for a new item and delivery target.
- 10 trials per policy and swarm size, with a 3000-timestep horizon.
- **Independent variables:** robot policy and swarm size (8–128 robots).
- **Dependent variables:** items delivered, blocked moves, and delays.



A pickup-and-delivery trial with 48 robots on a Conflict-Free Directed Roadmap with a specialized depot roadmap. Robots are colored spheres; the depot cell is outlined in black; delivery cells match their robot's color; blocked robots' cells are outlined in red.

Results



Policy performance vs. swarm size. Median over 10 trials; shading is the interquartile range.

- Across policies, the swarm delivers more items as it grows, up to a point, after which congestion slowdowns outweigh parallelization and productivity falls.
- The *congestion policy* sustained productivity best at large swarm sizes.
- It also incurred the fewest blocked movements at every swarm size tested.

Conclusions and future work

Contributions

- An algorithm that deterministically generates a strongly-connected roadmap on the surface of any admissible 3D structure for collective-construction pathfinding.
- A distributed, multi-agent pathfinding strategy that relies only on a Conflict-Free Directed Roadmap and a local occupancy rule to ensure collision-free trajectories.
- A comparison of robot policies for navigating a congested roadmap.

Future work

- Apply our proposed strategy to multi-agent collective construction in simulation.
- Adapt the roadmap for the ARMADAS robotic construction system [1].

References

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